

OSI Model

Open Systems Interconnection — 7-Layer Reference Framework

The OSI (Open Systems Interconnection) model is a seven-layer conceptual framework developed by ISO to standardize how data is transmitted across networks. Each layer has a specific role and communicates with the layers directly above and below it.

#	Layer	Function	Key Elements
7	Application	Provides network services to applications	HTTP, FTP, SMTP, DNS
6	Presentation	Data formatting, encryption & compression	TLS/SSL, JPEG, MPEG
5	Session	Session setup, maintenance & termination	Checkpoints, dialog control
4	Transport	End-to-end delivery, segmentation & ports	TCP, UDP
3	Network	Routing & logical (IP) addressing	Routers, packets, IP
2	Data Link	Node-to-node delivery via MAC addresses	Switches, bridges, LLC/MAC
1	Physical	Raw bit transmission over physical medium	Hubs, cables, repeaters

Layer Details

Layer 1 – Physical Layer

Handles the actual physical transmission of raw bits over a medium. Responsible for bit synchronization, bit rate control, and transmission modes. Devices include hubs, repeaters, and cables.

Layer 2 – Data Link Layer

Manages node-to-node delivery using MAC addresses. Divided into LLC (error/flow control) and MAC (medium access) sublayers. Devices include switches and bridges.

Layer 3 – Network Layer

Handles routing and logical (IP) addressing to transmit data across different networks. Devices include routers. Data units are called packets.

Layer 4 – Transport Layer

Responsible for end-to-end delivery, segmentation, reassembly, and port addressing. Supports connection-oriented (TCP) and connectionless (UDP) services.

Layer 5 – Session Layer

Manages session establishment, maintenance, and termination. Provides synchronization checkpoints and dialog control.

Layer 6 – Presentation Layer

Handles data formatting, encryption/decryption, and compression. Protocols include TLS/SSL; formats include JPEG and MPEG.

Layer 7 – Application Layer

The top layer, closest to the user. Provides network services to applications via protocols such as HTTP, FTP, SMTP, and DNS.

Data Flow: Encapsulation & Decapsulation

Data travels **down** all seven layers on the sender's side (**encapsulation**) — each layer adds its own header information — and back **up** on the receiver's side (**decapsulation**), where each layer strips its corresponding header until the original data reaches the application.

Understanding the OSI model is fundamental to networking, as it provides a universal framework for diagnosing issues, designing systems, and ensuring compatibility across different networking technologies and vendors.